Subject: Bio303 (Notes)

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VU Medical Zone (biotechnologists)

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Bio303 objective notes for mid term

1) Living cells and organisms must perform work to stay.
    a) Alive
    b) to grow
    c) to reproduce
    d) all
2) Animals also convert the chemical energy of fuels into the _____.
    a) motion
    b) heat
    c) Both a and b
    d) Non
3) Define Bioenergetics?
    Bioenergetics is the quantitative study of theenergy transductions changes of one form of energy into another—that occur in living cells,and of the nature and function of the chemical processes underlying these transductions.
4) The unit of G and H is/are _____.
    a) joules/mole
    b) calories/mole
    c) joules/Calvin
    d) A and b
    e) all of the above
5) 1 Cal. is equal to _____Joules.
    a) 4.127
    b) 4.157
    c) 4.184
    d) 4.814
6) The term "entropy," which literally means "achange within," was first used in _____ by Rudolf Clausius
    a) 1850
    b) 1851
    c) 1852
    d) 1851
7) Entropy, S is a _____expression for therandomness or disorder in a system.
    a) Quantitative

b) Qualitative
c) Competitive
d) Both and b

8) The reactions of photosynthesis take place in _____stages.
a) One
b) Two
c) Three
d) Four

9) Define Phosphorylationpotential?
The actual free energy of hydrolysis of ATP under intracellular conditions is often called its phosphorylation potential, ΔGp.

10) Which of the following are not the Phosphorylated compounds?
a) ADP
b) ATP
c) AMP
d) DTP

11) Acyl transfer: acyl group transfer generally involves The addition of a nucleophile to the carbonyl carbon of an acryl group to form a _____
a) Dihedral intermediate
b) Trihedral intermediate
c) tetrahedral intermediate
d) pentahedral intermediate

12) Carbone atoms can exist in _____ oxidation status, depending on the elements with which they share electrons
a) Two
b) Three
c) Four
d) Five

13) In some biological oxidation (but not all), a carbon atom becomes covalently bonded to an oxygen atom (a enzyme catalyze these reaction calls _____.
a) Oxygenizes
b) Oxidases
c) Catalases

d) Incubator

14) _____is the loss of electron or an increase in oxidation state by a molecule, atom, or ion.

a) Oxidation
b) Reduction
c) Oxygenases
d) Non

15) Phosphorous can form _____covalent bonds.

a) Three
b) Four
c) Five
d) Six

16) Glycosyl group Transfers involves nucleophilic substitution at _____of a sugar ring.

a) C-1
b) C-2
c) C-3
d) C-4

17) Write down the three major class of reaction in which C-C bonds are formed or broken?

(1)Aldol condensation
(2) Claisen condensation
(3) Decarboxylation

18) Most cells have the capacity to carry out _____of specific, enzyme-catalyzed reactions.

a) A few
b) Hundreds
c) thousands
d) lakes

19) The total free energy of the system (reactants + products) always becomes more _____as the reaction proceeds.

a) Positive
b) Negative
c) No change
d) Sometimes positive and sometimes negative

20) The free-energy change for a reaction is _____of the pathway by which the reaction occurs.
a) Independent
b) Dependent
c) Non
d) Both

21) ∆ G of any reaction proceeding spontaneously toward its equilibrium is always negative becomes _____as the reaction proceeds, and is zero at the point of equilibrium.
a) More negative
b) less negative
c) positive
d) non

22) If the equilibrium constant for a given chemical reaction is 1.0, the standard free-energy change of that reaction is _____.
a) 0.0
b) 0.5
c) 1
d) 2

23) When ∆ G is large and negative, the reaction tends to go in the _____direction.
a) Revers
b) Forward
c) Sometimes reveres and Sometimes forward
d) None of the above

1. In the phosphocreatine, the_____ bond can be hydrolyzed to generate free creatine and Pi.
a) P-N
b) C-N
c) C-O
d) C-P

24) The release of Pi and the resonance stabilization of creatine favor the _____reaction.
a) Forward
b) Revers

c) Equilibrium constant
d) Non

25) The standard free energy change of phosphocreatine is large and negative (ΔG' 0 =?
a) -49,3 kj/mol)
b) +49,3 kj/mol)
c) +94,3 kj/mol)
d) -94,3 kj/mol)

26) 1,3-bisphosphoglycerate contains an _____bond between the carboxyl group at C-1 and phosphoric acid.
a) Covalent
b) Anhydride
c) Hydrogen
d) None of these

27) High-energy‘ compounds have a ΔG' 0 of hydrolysis more negative than_____;
a) -25 kj/mol
b) -15 kj/mol
c) -10 kj/mol
d) -20 kj/mol

28) _____serves as the universal energy currency in all living cells.
a) ATP
b) ADP
c) Both and b
d) GTP

29) Which ofthe following processes do involve direct hydrolysis of ATP (or GTP)?
a) Muscle contraction
b) The movement of enzymes along the DNA
c) The movement of ribosomes along the RNA
d) All of the above

30) ATP: although thermodynamically unstable, it is kinetically stable _____kJ/mol ◊ ATP does not spontaneously donate phosphoryl group to water
a) 200 to 400

b) 300 to 400
c) 400 to 600
d) 400 to 800

31) The contribution of ATP to a reaction is often shown as a _____, but is almost always a _____ process.
a) two-step , single step
b) single step ,two-step
c) two-step , three step
d) three step, two-step

32) The phosphoryl group is displaced by _____and released as Pi.
a) NH2
b) NH3
c) NH4
d) All

33) The nucleophile may be an alcohol _____.
a) (ROH),
b) (RCOO),
c) a phosphoanhydride
d) All of the above

34) Hydrolysis of a-b and a-g phosphoanhydride bonds respectively are?
a) 46,36 kJ/mol
b) 46,31 kJ/mol
c) 40,33 kJ/mol
d) 40,46 kJ/mol

35) AMP transfer from ATP to carboxyl group of FA ◊ Form fatty acyl adenylate and liberating PPi: exergonic has ΔG 'o = _____kJ/mol
a) 40
b) 44
c) -45
d) +45

36) _____ is a coenzyme that reversibly binds to enzymes and _____is a prosthetic group that remains tightly bound at the active site of an enzyme.
a) NAD+ , FAD
b) NADPH , FADP

c) NAD+ , FAD+
d) NAD+ , FADB

37) ATP bind with head of _____ and dissociate from _____.
a) Actin , Myosin
b) Myosin, Actin
c) Mylin , Actin
d) Non

38) Define Ping-pong mechanism of nucleoside diphosphate kinase?
The enzyme binds its first substrate (ATP in our example), and a phosphoryl group is transferred to the side chain of a His residue. ADP departs, and another nucleoside (or deoxynucleoside) diphosphate replaces it, and this is converted to the corresponding triphosphate by transfer of the phosphoryl group from the phosphohistidine residue.

39) Define reductant and oxidant?
The electron-donating molecule in an oxidation-reduction reaction is called the reducing agent or reductant; the electron accepting molecule is the oxidizing agent or oxidant. These are combinedly also called Conjugate redox pair.

40) The neutral term reducing equivalent is commonly used to designate a single _____equivalent participating in an oxidation-reduction reaction.
a) Proton
b) Electron
c) Neutron
d) All

41) A salt bridge containing a saturated _____solution provides a path for counter-ion movement between the test cell and the reference cell.
a) KCl
b) NACL
c) MgCl
d) CFCl

42) Total concentration of NAD+ + NADH in most tissues are_____ M.
a) 10 power -4

b) 10 power -5
c) 10 power -6
d) 10 power -8

43) Total concentration of NADP+ + NADPH in most tissues are _____M
a) 10 power -4
b) 10 power -5
c) 10 power -6
d) 10 power -8

44) More than _____ enzymes are known to catalyze reactions in which NAD+ (or NADP+) accepts a hydride ion from a reduced substrate or NADPH (or NADH) donates a hydride ion to an oxidized substrate.
a) 200
b) 250
c) 300
d) 400

45) What are Cryptochromes?
Cryptochromes are a family of flavoproteins, widely distributed in the eukaryotic phyla that mediate the effects of blue light on plant development and the effects of light on mammalian circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period).

46) Photolyases use the energy of absorbed light to repair chemical defects in_____.
a) RNA
b) DNA
c) Protines
d) Non

47) Fully reduced and partially reducedflavin nucleotides for oxidation reduction have lengths of respectively_____.
a) 360 and 450 nm
b) 370 and 440 nm
c) 350 and 450 nm
d) 400 and 450 nm

48) The energy gain comes in the *payoff phase* of _____.

a) Glycolysis
b) Oxygenasis
c) Glyoxygensis
d) Non

49) Each molecule of glyceraldehyde 3-phosphate is oxidized and phosphorylated by inorganic phosphate (*not* by ATP) to form_____.

a) 1, 3-bisphosphoglycerate
b) 1, 4-bisphosphoglycerate
c) 1, 5-bisphosphoglycerate
d) 1, 6-bisphosphoglycerate

50) Much of this energy is conserved by the coupled phosphorylation of _____molecules of ADP to ATP.

a) Two
b) Three
c) Four
d) Five

51) Pyruvate is the end product of the second phase of_____.

a) Glycolysis
b) Hydrolysis
c) Oxygenasis
d) Glyoxygensis

52) For each glucose molecule, _____ATP are consumed in the preparatory phase and _____ATP are produced in the payoff phase,

a) Two, one
b) Two, two
c) Two, four
d) Two, five

53) Each phosphoryl group, represented as P, has two _____charges.

a) Negative
b) Positive

54) Pyruvate is oxidized, with loss of its carboxyl group as _____.

a) CO2

b) CO
c) CHO
d) ClO

55) The second route for pyruvate is its reduction to lactate via _____fermentation
a) Acetic acid
b) lactic acid
c) pyloric acid
d) pharmic acid

56) The third major route of pyruvate catabolism leads to formation of_____.
a) Methanol
b) Ethanol
c) Hexanol
d) Non

57) In some plant tissues and in certain invertebrates, protists, and microorganisms such as brewer‘s or baker‘s yeast, pyruvate is converted under _____conditions to ethanol and CO2, a process called ethanol (alcohol) fermentation.
a) hypoxic
b) anaerobic
c) both a and b
d) non of these

58) For each molecule of glucose degraded to pyruvate, _____molecules of ATP are generated from ADP and Pi.
a) One
b) Two
c) Three
d) Four

59) Write down the two processes equation of glycolysis?

(1) The conversion of glucose into pyruvate is exergonic:

Glucose + 2NAD+ 2 pyruvate + 2NADH + 2H+ ,ΔG°'1 = -146 kJ/mol (Eqn II)

(2) and the formation of ATP from ADP and Pi is endergonic:

2ADP + 2Pi 2ATP + 2H2O , ΔG°'2 = 2(30.5 kJ/mol) = 61 kJ/mol (Eqn III)

60) The phosphate groups are ionized at pH_____, thus giving each of the intermediates of glycolysis a net negative charge.
a) 5
b) 6
c) 7
d) 9

61) Binding of phosphate groups to the active sites of enzymes provides binding energy that contributes to _____.
a) lowering the activation energy
b) increasing the specificity of enzyme-catalyzed reactions
c) Both
d) Non

62) In_____, Arthur Harden and William Young tested their hypothesis that inhibitors of proteolytic enzymes would stabilize the glucose fermenting enzymes in yeast extract.
a) 1960
b) 1906
c) 1966
d) 1956

63) Harden and Young soon discovered that glucose added to their yeast extract was converted to a hexose bisphosphate (the ―Harden- Young ester,‖ eventually identified as_____.
a) fructose 1,3-bisphosphate
b) fructose 1,6-bisphosphate
c) glucose 1,3-bisphosphate
d) glucose 1,6-bisphosphate

64) In the first step of glycolysis, glucose is activated for subsequent reactions by its phosphorylation at _____to yield glucose 6-phosphate, with ATP as the phosphoryl donor.
a) C-4
b) C-5
c) C-6
d) C-7]

65) Kinesis are _____.

Kinases are a subclass of transferases.

66) The acceptor in the case of hexokinase is a hexose, normally_____, although hexokinase also catalyzes the

phosphorylation of other common hexoses, such as D-fructose and D-mannose, in some tissues.

a) D-glucose
b) Beta-glucose
c) Alpha-glucose
d) A and b

67) Hexokinase, like many other kinases, requires _____for its activity, because the true substrate of the enzyme is not ATP-4 but the MgATP-2 complex

a) Mg^{+2}
b) $MgCl^{+2}$
c) Mg^{+4}
d) $Mg^{+2}Cl^{+2}$

68) Define isozymes?

Two or more enzymes that catalyze the same reaction but are encoded by different genes are called isozymes

69) Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named _____.

a) *Bisphosphates*
b) *Tryphosphates*
c) *Tetreaphosphates*
d) *Pantaphosphates*

70) The PFK-1 reaction is essentially _____under cellular conditions.

a) Irreversible
b) Reversible
c) Both
d) Non

71) cell's ATP supply is depleted or when the ATP breakdown products_____.

a) ADP
b) AMP
c) ADPH
d) Both a and b

72) The enzyme is inhibited whenever the cell has ample ATP and is well supplied by other fuels such as_____.
a) Glucose
b) fatty acids
c) lipds
d) non

73)

74) The enzyme fructose 1, 6-bisphosphate aldolase, often called simply _____a reversible aldol condensation.
a) Aldolase
b) Catalyzes
c) Anaphlase
d) Both a and b

75) The other product, **dihydroxyacetone phosphate**, is rapidly and reversibly converted to glyceraldehyde 3-phosphate by the fifth enzyme of the glycolytic sequence_____.
a) **biose phosphate isomerase**
b) **triose phosphate isomerase**
c) **tetrose phosphate isomerase**
d) **pantose phosphate isomerase**

76) After the triose phosphate isomerase reaction, the carbon atoms derived from _____of the starting glucose are chemically indistinguishable.
a) C-1
b) C-2
c) C-3
d) All of the above

77) The hexose molecule has been phosphorylated at C-1 and C-6 and then cleaved to form _____molecules of glyceraldehyde 3-phosphate.
a) Two
b) Three
c) Four
d) Five

78) The payoff phase of glycolysis includes the energy-conserving phosphorylation steps in which some of the chemical energy of the glucose molecule is conserved in the form of_____.
a) ATP
b) NADH
c) ADP
d) Both a and b

79) The conversion of two molecules of glyceraldehyde 3-phosphate to two molecules of pyruvate is accompanied by the formation of four molecules of_____.
a) ATP
b) ADP
c) NADH
d) Both a and b

80) Ethanol and CO2 are the end products of _____fermentation .
a) Ethanol
b) Mathanol
c) Hexanol
d) Pantanol

81) Yeast and other microorganisms ferment glucose to_____, rather than to lactate.
a) ethanol
b) CO2
c) Glycogen
d) Both A and B

82) Weite the final eqution of ethanol fermentation?
Glucose + 2ADP + 2Pi ⟶ 2 ethanol + 2CO2 + 2ATP + 2H2O

83) _____is converted to pyruvate by glycolysis, and the pyruvate is converted to ethanol and CO2 in a two-step process.
a) Lipids
b) Glucose
c) Protiens
d) Non

84) Acetaldehyde is then reduced to ethanol under _____conditions.
a) Anaerobic
b) Aerobic
c) Both
d) Non

85) In lactic acid fermentation, there is no net change in the ratio of _____atoms.
a) nitrogen to carbon
b) oxygen to carbon
c) hydrogen to carbon
d) hydrogen to oxigen

86) What is the meaning of Lactobacillus?
a) Milk

b) Rod
c) sphere
d) Both a and

87) Lactobacillus is found over ____different species.
a) 25
b) 50
c) 75
d) 100

88) _____Found on plants and in the digestive system of animals such as cows and humans.
a) Lactobacillus
b) Lactobacterea
c) Lactobfebri
d) Lactostomata

89) What is the meaning of Lactococcus?
a) Milk
b) Rod
c) sphere
d) Both a and c

90) Write some traditional spontaneously fermented milks contain species that can reside in the human digestive tract?
Lactobacillus *fermentum*,
L. *casei*
L *brevis*
L *plantarum* (from picked vegetables)
L *acidophilus*

91) Fermentations Are Used to Produce Some _____.
a) Common Foods
b) Industrial Chemicals
c) Both
d) Non

92) Yogurt, is produced when the bacterium *Lactobacillus bulgaricus* ferments the _____in milk, producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt.
a) Lipids
b) Carbohydrate

c) Glucose
d) Non

93) How cheese produced?
Cheese is produced by using *Propionibacterium freudenreichii*, which ferments milk to produce *propionic acid and CO2*; the propionic acid precipitates milk proteins, and bubbles of CO2 cause the holes characteristic of Swiss Cheese.\

94) The silage that results from the _____can be kept as animal feed for long periods without spoilage.
a) Agricultural fermentation
b) Industrial fermentation
c) Domastic fermentation
d) Non of these

95) Fermentation is derived from the Latin Verb _____meaning to boil.
a) Fares
b) Fevere
c) Farmena
d) Farmenta

96) Decribe and berify the meaning of fermentaion biochemistry and industrial microbiologists?
Biochemical meaning relates to the generation of energy by the catabolism of organic compounds
Whereas meaning in industrial microbiology tends to be much broader. Industrial microbiologists have extended the term - fermentation - to describe: —Any process for the production of product by the mass culture of microorganism.

97) Write down some requirements of fermentation process.
i. The formulation of media
ii. The sterilization of the medium
iii. The production of an active, pure culture in sufficient quantity to inoculate the production vessel.
iv. The growth of the organism
v. The extraction of the product and its purification
vi. The disposal of effluents produced by the process.

98) Any industrial fermentation operation can be broken down into _____main stages.
a) Two
b) Three
c) Four

d) Five

99) Mention the names of three main stages of the industrial fermentation?

1. upstream processing

2. the fermentation process

3. Downstream processing

100) _____includes *formulation of the fermentation medium*, *sterilisation* of air, fermentation medium and the fermenter, *inoculum preparation* and *inoculation* of the medium.

a) upstream processing

b) the fermentation process

c) Downstream processing.

d) Non of these

101) _____involves the *propagation of the microorganism* and *production of the desired product*. The fermentation process can be categorised depending on various parameters.

a) upstream processing

b) the fermentation process

c) Downstream processing

d) Non of these

102) includes the recovery of the products in a pure state and the effluent treatment.

a) upstream processing

b) the fermentation process

c) Downstream processing

d) Non of these

103) Product recovery is carried out through a series of operations including *cell separation* by _____.

a) Settling

b) centrifugation

c) Filtration

d) All of the above

104) *Product recovery* by disruption of cells (if the product is produced intracellularly); *extraction* and *purification* of the product. Finally, the *effluents are treated* by _____methods.

a) Chemical

b) physical

c) Biological

d) All of the above

105) A medium which is used for a large scale fermentation, in order to ensure the sustainability of the operation, should have the all the characteristics except?

a) It should be cheap and easily available

b) It should maximize the growth of the microorganism, productivity and the rate of formation of the desired product

c) It should minimize the formation of undesired products

d) It should be highly endothermic

106) Fermentation technique can be divided into_____.

a) surface (emersion) techniques

b) submersion techniques

c) Both and b

d) Non

107) In the surface techniques, the microorganisms are cultivated on the_____.

a) surface of a liquid

b) solid substrate

c) both a and b

d) non

108) In the submersion processes, the microorganisms grow in a_____.

a) liquid medium

b) solid medium

c) jelly medium

d) non of these

109) Most processes are_____, and for these the medium must be vigorously aerated.

a) Aerobic

b) Anaerobic

c) Both a and b

110) All important industrial processes (production of biomass and protein, antibiotics, enzymes and sewage treatment) are examples of what?

a) Surface techniques
b) Submersion processes
c) Non

111) MO (microrganisms) require_____ for media.
a) nutrients and salinity
b) O2 requirement
c) Temp and pH
d) All of the above

112) How many kinds of media?
a) Two
b) Three
c) Four
d) Five

113) Which Factors to be consider when designing a fermenter?
Aseptic and regulatory capability, long-term reliability
Adequate aeration and agitation
Low power consumption
Temperature and pH controls
Sampling facilities

114) The fermenter needs to be able to control such factors as _____.
a) Temperature
b) pH
c) dissolved oxygen levels
d) all of the above

115) Which of the following is icliuded in aeration and Agitation in the fermenter?
a) Shake culture
b) Stirrers
c) Baffles
d) All of the above

116) A biorector is a device in which the organisms are cultivated and motivated to form a _____.
a) desired product
b) normal product
c) traditioal product

d) non

117) Fermenter: for microbes/ Bioreactor : for eukaryotic cells, size variable ranging from _____million litres or more.

a) 20-200
b) 25-250
c) 20-250
d) Non

118) The only material added and removed during the course of a batch fermentation is the_____.

a) gas exchange
b) pH control solutions
c) both a and b
d) non

119) *Aspergillus niger* and *Lactobacillus Delbruckii* are the microbes used to commercially produce _____.

a) citric acid
b) lactic acid
c) Propanoic acid
d) Both a and b

120) Mention the phases of batch cultures?

- lag phase
- lag /exponential growth phase
- stationary phase
- death phase

121) The phase in the fermentation process. The cells have adjusted to their environment and rapid growth takes place, cell growth rate is highest in this phase. This is called_____.

a) lag phase
b) exponential growth
c) stationary phase
d) death phase

122) Define and explain the Continuous fermentation shortly?

A Continuous fermentation is a process in which fresh medium (substrate) is continuously added in the bioreactor, and biomass or products containing left over nutrients and microorganisms are continuously removed at the same rate to keep the culture volume constant.

123) Which of the following is not the Advantage of Fermentation?

a) Preserves and enriches food, improves digestibility, and enhances the taste and flavour of foods.

b) Potential of enhancing food safety by controlling the growth and multiplication of a number of pathogens in foods.

c) Important contribution to human nutrition, particularly in developing countries, where economic problems pose a major barrier to ensuring food safety.

d) Preservation and detoxification of the food

124) Write down the Disadvantages of Fermentation?

- Low energy consumption due to the mild operating conditions relatively low capital and operating costs relatively simple technologies.
- They cause highly specific and controlled changes to foods by using enzymes.
- Preservation and detoxification of the food.
- Waste treatment.

125) Commercially important products of fermentation can be described in _____major groups.

a) Three
b) Four
c) Five
d) Six

126) Commercially important products of fermentation can be described in five major groups, enlist them?

1. Biomass
2. Primary metabolites
3. Bioconvertion or biotransformation products
4. Enzymes
5. Recombinant products

127) Vaccines, hormones such as insulin and growth hormones etc belongs which type of product?

a) Primary metabolites
b) Bioconvertion or biotransformation products
c) Enzymes
d) Recombinant products

128) Which of the following is not the organic acid?
a) Glutamic acid
b) Citric acid
c) Acetic acid
d) Lactic acid

129) Define Gluconeogenesis?
The process of synthesis of glucose from non-carbohydrate sources is known as Gluconeogenesis (—new formation of sugarǁ), OR Gluconeogenesis is the pathway, which converts pyruvate and related three- and four-carbon compounds to glucose.

130) For the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the _____is the sole or major fuel source.
a) Muscles cells
b) Blood
c) Food
d) Non

131) The brain alone requires about_____g of glucose each day.
a) 110
b) 120
c) 130
d) 140

132) More than half of all the glucose stored as glycogen in _____.
a) muscle
b) liver
c) Spleen
d) Both a and b

133) Which of the following Organism do Gluconeogenesis?
a) all animals
b) plants
c) fungi and microorganisms
d) All of the above

134) Which of the organ is doing Gluconeogenesis?
a) in the liver
b) To a lesser extent in renal cortex
c) in the epithelial cells that line the inside of the small intestine

d) All of the above

135) Write down the Pathways involved in Gluconeogenesis?

Reverse glycolysis

Tri carbocyclic acid cycle (TCA) or Kreb's cycle

Cori cycle

Glucose-alanine cycle

136) Write down the biomedical importance of Gluconeogenesis?

- To maintain the blood glucose concentration during prolonged fasting or starvation when sufficient carbohydrate is not available from the diet or glycogen reserves.
- It maintains the level of intermediates of the TCA cycle even when fatty acids are the main source of acetyl coA in the tissues.
- It clears the lactate produced by muscle and erythrocytes and glycerol produced by adipose tissue.

137) Define Cori cycle?

The pathway through which lactate produced by anaerobic glycolysis in skeletal muscle returns to the liver and is converted to glucose, which moves back to muscle and is converted to glycogen is called the Cori cycle

138) The phosphorylation of pyruvate is achieved by a roundabout sequence of reactions that in eukaryotes requires enzymes in the_____.

a) cytosol
b) mitochondria
c) Liver
d) Both a and b

139) The CO2 incorporated in the pyruvate carboxylase reaction is lost here as_____.

a) CO

b) CO_2
c) CO_3
d) CO_4

140) _____is the first regulatory enzyme in the gluconeogenic pathway.
a) Pyruvate carboxylase
b) Lactate carboxylase
c) Glycerol carboxylase
d) Non

141) In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by_____.
a) PEP carboxykinase
b) Pyruvate carboxylase
c) Lactate carboxylase
d) Glycerol carboxylase

142) Fructose_____ allosterically activates the glycolysis enzyme Phosphofructokinase.
a) 1,3-bisphosphate
b) 2,4-bisphosphate
c) 2,6-bisphosphate
d) 3,6-bisphosphate

143) Fructose-2, 6-bisphosphate stimulates_____.
a) Nitrogen cycle
b) Glycolysis
c) Glucose formation
d) Non

144) Fructose-2, 6-bisphosphate also activates _____of the gene for Glucokinase.
a) Transcription
b) Transfer
c) Removal
d) Introduction

145) Fructose-2, 6-bisphosphate _____the relaxed state, activating Phosphofructokinase even at high [ATP].
a) Promotes

b) Starts
c) Stop
d) Favor

146) Thus activation by fructose-2, 6-bisphosphate, whose concentration fluctuates in response to _____signals, supersedes local control by [ATP].

a) external hormonal
b) internal hormonal
c) external enzymes
d) internal enzymes

147) Write down the effects of glucagon-cAMP cascade in liver?

Gluconeogenesis is stimulated.
Glycolysis is inhibited.
Glycogen breakdown is stimulated.
Glycogen synthesis is inhibited.
Free glucose is formed for release to the blood

148) Under what conditions the Glycolysis slows and Gluconeogenesis increases?

Glycolysis slows because fructose-2, 6-bisphosphate is not available to activate Phosphofructokinase. Gluconeogenesis increases because of the decreased concentration of fructose-2, 6-bisphosphate, which would otherwise inhibit the gluconeogenesis enzyme Fructose-1, 6-bisphosphatase.

149) The biosynthetic pathway to glucose described so far allows the net synthesis of glucose not only from pyruvate but also from the _____carbon intermediates of the citric acid cycle.

a) Four
b) Five
c) Six
d) All of the above

150) Some or all of the carbon atoms of most amino acids derived from _____are ultimately catabolized to pyruvate or to intermediates of the citric acid cycle.

a) Proteins
b) Lipids
c) Glucose
d) Minerals

151) Alanine and glutamine, the principal molecules that transport _____from extrahepatic tissues to the liver are particularly important glucogenic amino acids in mammals.

a) Phosphate group
b) amino groups
c) hydroxyl group
d) all of the above

152) No net conversion of fatty acids to glucose occurs in_____.

a) Mammals
b) Birds
c) Fishes
d) Non

153) The catabolism of most fatty acids yields _____.

a) acetyl-CoA and KCL
b) acetyl-CoA and glucose
c) acetyl-CoA and ATP
d) only acetyl-CoA

154) The *oxidative* pentose phosphate pathway (phosphogluconate pathway or hexose monophosphate pathway) brings about oxidation and decarboxylation of glucose 6-phosphate (G6P) at_____, reducing NADP+ to NADPH and producing pentose phosphates.

a) C-1
b) C-2
c) C-3
d) C-4

155) In most animal tissues, the major catabolic fate of glucose 6-phosphate is glycolytic breakdown to pyruvate, much of which is then oxidized via the citric acid cycle, leading to_____.

a) ATP formation
b) ADP formation
c) ADPH formation
d) None

156) Oxidation of glucose 6-phosphate to pentose phosphates by the pentose phosphate pathway. There are _____phases of the pentose phosphate pathway.

a) Two

b) Three
c) Four
d) Five

157) Pentose Phosphate Pathway NADPH formed in the oxidative phase is used to reduce_____.
a) Glutathione
b) GSSG (Glutathione disulfide)
c) to support reductive biosynthesis
d) All of the above

158) The other product of the oxidative phase is ribose 5-phosphate, which serves as a precursor for_____.
a) Nucleotides
b) Coenzymes
c) nucleic acids
d) all of the above

159) The Oxidative Phase Produces _____.
a) Pentose Phosphates
b) NADPH
c) Both a and b
d) Non

160) The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific enzyme _____.
a) Lactonase
b) Lactokinase
c) Lactoplase
d) Lactopays

161) The net result is the production of NADPH is+_____.
a) A reducing agent for biosynthetic reactions
b) Ribose 5-phosphate
c) A precursor for nucleotide synthesis
d) All of the above

162) The genetic master plan is contained in the nucleotide sequence of_____.
a) RNA
b) DNA

c) Proteins
d) Mitochondria

163) RNA is a polymer of ribonucleotides of _____.
a) Adenine, Uracil, Guanine and Cytosine
b) Adenine, thiamine, Guanine and Cytosine
c) Adenine, Uracil, Guanine, thiamine and Cytosine
d) None

164) RNA is a polymer of ribonucleotides of Adenine, Uracil, Guanine and Cytosine, joined together by 3´-5´_____.
a) Phosphodiester bonds
b) Glycosidic bonds
c) Covalent bond
d) Phosphate bond

165) The pentose sugar of RNA is _____.
a) D-ribose
b) D-oxyribose
c) D-fructose
d) Both a and b

166) RNA is found in the_____.
a) nucleolus
b) ribosomes
c) mitochondria and cytoplasm
d) all of the above

167) The genetic material for some animal and plant viruses is _____rather than DNA.
a) Mitochondria
b) RNA
c) Only sugars
d) None

168) Write and define types of RNA?
There are three types of RNA. These are:
Messenger RNAs (mRNAs)
Ribosomal RNAs (rRNAs)
Transfer RNAs (tRNAs)
• messenger RNAs (mRNAs) - transfer genetic information from DNA to the protein-synthesizing machinery.

• ribosomal RNAs (rRNAs) - contribute to the formation and function of ribosomes
• transfer RNAs (tRNAs) - adapter molecules that carry specific amino acids for protein synthesis

169) Small nuclear RNA (snRNA)- play pivotal roles in RNA processing, particularly _____.
a) rRNA processing
b) tRNA processing
c) mRNA processing
d) DNA processing

170) Some RNA molecules have intrinsic catalytic activity these RNA enzymes, are called_____.
a) Ribozymes
b) Intrinsic enzymes
c) Riboenzymes
d) None

171) RNA are considerably _____ DNA.
a) Smaller than
b) Larger than
c) Equal to
d) None

172) Thymine is present in the _____.
a) rare case of tRNA
b) RNA
c) DNA
d) Both A and C

173) However, given the proper complementary base sequence with opposite polarity, the _____is capable of self- folding like a hairpin.
a) single strand of RNA
b) any one strand or DNA
c) DNA and RNA
d) Non

174) Hydrolysis: RNA can be hydrolyzed by alkali to 2',3' cyclic diesters of the mononucleotides, compounds that cannot be formed from alkali-treated DNA because of the absence of a_____.
a) 2'-hydroxyl group
b) Hydrogen group

c) Carboxyl group
d) None

175) DNA forms RNA by transcription whereas the process by which RNA form DNA is called_____.
a) reverse transcription
b) transcription
c) translation
d) none

176) _____structure involves coil formation of the polyribonucleotide chain.
a) Secondary
b) Primary
c) Secondary
d) Tratioray

177) Define and explain Tertiary Structure of RNA?
It is the folding of the molecule into three dimensional structure. Cross-linking at various sites stabilized by hydrophobic and hydrogen bonds produces a compactly coiled globular structure.

178) mRNA size arrange from--------- nucleotides.
a) 500-4000
b) 500-6000
c) 500-60000
d) 600-60000

179) mRNA is most heterogeneous in----------.
A) Abundance
B) size
C) base sequence and Stability
D) all of the above

180) mRNA comprise about ----------% of the total cellular RNA.
a) 1
b) 2
c) C) 2.5
d) D) 10

181) --------- is the smallest of the three major species of RNAs.
a) mRNAA)

b) rRNA
c) tRNA
d) none

182) Extensive ---------- base pairing which leads to a characteristic CLOVER-LEAF structure.
a) inter chain
b) intra chain
c) interior chain
d) none

183) --------- remain largely in cytoplasm.
a) mRNA
b) rRNA
c) tRNA
d) none

184) The folds in tRNA are stabilized by ---------- between complementary bases of the same strand.
a) nitrogen bonds
b) hydrogen bonds
c) hydroxide bonds
d) phosphate bonds

185) tRNAs compose roughly --------of total cellular RNA.
a) 2%
b) 12%
c) 20%
d) 30%

186) There are at least ---------- species of tRNA molecules in every cell.
a) 20
b) 200
c) 2000
d) 20000

187) tRNA molecules consist of ---------- nucleotides in a particular sequence.
a) 64-74
b) 74-84
c) 80-90

d) 74-95

188) All tRNA molecules contain ------ main arms or loops.
a) 3
b) 4
c) 5
d) 6

189) The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, uracil but also contain unusual bases. These unusual bases also called ----------.
a) emerging bases
b) undefined bases
c) modified bases
d) extra bases

190) The modified bases include which of the following.
a) Dihydrouracil
b) Pseudouridine
c) Thymine
d) all of the above

191) ---------- end of mRNA is called poly a tail.
a) 3
b) 5
c) Both
d) none

192) ------- contains thymine, pseudouridine and cytosine.
a) 3-D arm
b) 1-Acceptor arm
c) 4-TΨC arm
d) none

193) Write down the function of mRNA?
The members of this class function as messengers to convey the information in a gene to the protein synthesizing machinery. The mRNA carries genetic information from the nuclear DNA to the cytosol, where it is used as a template for protein synthesis.

194) Define and berify the 1-Acceptor arm of tRNA?

1-Acceptor arm: This is made up of unpaired sequences of cytosine-cytosine-adenine (CCA) at the 3´end. The 3´OH group of adenine binds with the carboxylic group of a specific amino acid and carries it to ribosomes for protein synthesis.

195) _____found in association with several proteins as a component of ribosomes.
a) tRNA
b) mRNA
c) rRNA
d) none

196) RNAs make up _____of the total RNA in the cell.
a) 30%
b) 50%
c) 60%
d) 80%

197) The ribosomal subunits are defined according to their _____in Svedberg units.
a) Size
b) Mass
c) Volume
d) sedimentation velocity

198) Svedberg unit is related to the _____of the compound.
a) molecular weight
b) shape
c) Molecular mass
d) Both a and b

199) The mammalian ribosome contains which of the following major nucleoprotein subunits.
a) 60S
b) 40S
c) Both a and b
d) None

200) The 60S subunit contains which of the folllwing one?
a) 5S rRNA
b) 5.8S rRNA
c) 28S rRNA

d) All of the above

201) There are how many specific polypeptides found in eukaryotic Ribosome.

a) 30
b) 40
c) 50
d) more than 50

202) The 40S subunit is smaller and contains which of the folllwing?

a) 5S rRNA
b) 5.8S rRNA
c) 18S rRNA
d) 28S rRNA

203) In eukaryotes, all of the ribosomal RNA molecules except the _____, which is independently transcribed, are processed from a single 45S precursor RNA molecule in the nucleolus- packed with the specific ribosomal proteins.

a) 5S rRNA
b) 5.8S rRNA
c) 18S rRNA
d) 28S rRNA

204) snRNA stands for which of the following?

a) Simple nuclear RNA
b) Small nuclear RNA
c) Super natural RNA
d) None

205) Small nuclear RNA (snRNA) are large number of small stable RNA species found in_____.

a) eukaryotic cells
b) prokaryotic cells
c) both a and a
d) none

206) Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells. Most of them are complexed with proteins to form _____.

a) Riboproteins

b) Nucleoproteins
c) Ribonucleoproteins
d) None of the above

207) Small nuclear RNA (snRNA) are distributed in the_____.
a) Nucleus
b) in the cytoplasm
c) ribosome
d) both a and b

208) NcRNAs exist in two general size classes write name and function?
Small consisting of microRNA (miRNAs) and silencing (siRNAs) and
Large consisting of long noncoding RNAs (lncRNAs)
The small ncRNAs termed microRNA (miRNAs) and silencing (siRNAs) typically inhibit gene expression at the level of specific protein production by

209) Lyases enzymes catalyze the addition of
a) NH_3
b) H_2O
c) CO_2 to double bonds
d) All of the above

210) IUB stands for what?
International Union of Biochemists

211) Ligases enzymes catalyze condensation reactions joining two molecules by forming.
a) C-O,
b) C-S,
c) C-N and C-C bonds
d) All

212) **Flavin nucleotides are tightly bound in _____.**

a) Flavoproteins

b) Lipoprotein

c) Lectoprotien

d) Non

213) **Dietary deficiency of Niacin, the vitamin form of NAD and NADP, causes _____.**

a)Pellagra

b) Ellipsy

c)Energy deficiency

d) All

214) In eukaryotes, all of the ribosomal RNA molecules except the _____, which is independently transcribed, are processed from a single 45S precursor RNA molecule in the nucleolus- packed with the specific ribosomal proteins.

e) 5S rRNA
f) 5.8S rRNA
g) 18S rRNA
h) 28S rRNA

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Bio303 objective notes for mid term